Runbook RB-112: Primary database failover

Owner: Platform Database Team · Version 3.2 · Last reviewed 2 October 2026

1. Purpose and scope

This runbook describes how to promote the standby PostgreSQL server when the primary in the Leeds data centre becomes unavailable. It applies to the order management cluster only. Other clusters use automatic failover and are covered by RB-090.

> **Before you start**
> Do not promote the standby while the primary is still accepting writes, because this creates two diverging timelines. If you are unsure, page the duty database engineer first.

2. Preconditions

1. The incident has been declared in the incident channel and an incident commander is named.
2. Replication lag on the standby is below 30 seconds, or the business has accepted the loss of recent orders.
3. You have a terminal on the bastion host with the dba-oncall role.

3. Procedure

3.1 Confirm that the primary is down

Check the replication status from the standby and note the last received log position in the incident record.

```
$ ssh dba@pg-orders-b.leeds.internal
$ sudo -u postgres psql -c "SELECT pg_last_wal_receive_lsn(), now() - pg_last_xact_replay_timestamp() AS lag;"
```

3.2 Fence the old primary

1. Remove the old primary from the load balancer pool:
   - 1.1 open the pool definition for orders-db-rw;
   - 1.2 set the old primary to drain and confirm that active connections fall to zero.
2. Stop the PostgreSQL service on the old primary if the host still responds.
3. Record the time of fencing in the incident timeline.

3.3 Promote the standby

Run the promotion on the standby host and wait until the server leaves recovery mode.

```
$ sudo -u postgres pg_ctl promote -D /var/lib/postgresql/16/main
$ sudo -u postgres psql -tAc "SELECT pg_is_in_recovery();"
f
```

3.4 Point the applications to the new primary

Update the service discovery record so that the write endpoint resolves to the promoted server. The change below is applied through the configuration repository and takes effect within two minutes.

```
# config/orders/database.yaml
write_endpoint:
  host: pg-orders-b.leeds.internal
  port: 5432
  pool_size: 40
read_endpoints:
  - host: pg-orders-c.york.internal
    port: 5432
```

4. Verification

- Place a test order in the staging tenant and confirm that it is stored.
- Check that the error rate on the order API has returned to its normal level.
- Rebuild a new standby from the promoted server within 24 hours.

5. Escalation contacts

Role	Team	Hours	Contact
Duty database engineer	Platform Database	24/7	pager group dba-oncall
Incident commander	Service Management	24/7	ext. 4400
Order management owner	Commerce Systems	08:00–18:00	ext. 4127
Network on call	Infrastructure	24/7	pager group net-oncall

6. Version history

Version	Date	Change
3.0	14 Mar 2026	Moved to PostgreSQL 16
3.1	9 Jun 2026	Added fencing step
3.2	2 Oct 2026	Updated contacts and YAML path